

# Claim-Gated Dynamic Service-Menu Optimization for Many-to-One Demand-Responsive Transit

Jin Ana<sup>a</sup>

<sup>a</sup>MoE Key Laboratory of Complex System Analysis and Management Decision, School of Economics and Management, Beihang University, Beijing, 100191, China

## ARTICLE INFO

### Keywords:

demand-responsive transit  
service-menu optimization  
meeting points  
pickup time windows  
multinomial logit  
paired replay  
artifact gates  
transportation operations

## ABSTRACT

Many-to-one demand-responsive transit (DRT) platforms increasingly face a service-design decision that is richer than route insertion alone. For each arriving request, the operator may display several feasible service bundles, where each bundle combines a pickup location, a pickup time-window treatment, and a price. This paper formulates that decision as dynamic service-menu optimization. The service bundle is written as  $b = (m, w, p)$ , with  $m$  denoting home or meeting-point pickup,  $w$  denoting pickup-window handling, and  $p$  denoting the displayed price. Passenger response is modeled experimentally with a multinomial-logit (MNL) choice model over displayed bundles and the outside option. The outside option represents refusal or lost demand, not accepted home pickup, so accepted home service, accepted meeting-point service, and opt-out are kept as separate accounting outcomes.

The manuscript follows a conditional diagnostic evidence path. It describes the formulation, the menu-construction and paired-replay evaluation pipeline, and the package-level claim gates that govern manuscript interpretation. In the current generated package, the claim-ready flag, strict claim-guard flag, and positive-claims permission flag are all false. Positive empirical claims about adaptive-menu effects, product ablations, adaptive-window increments, menu-construction value, exact/greedy computational credibility, and semi-real case evidence remain blocked by readiness and strict claim-guard status. The contribution is therefore a claim-gated service-menu optimization framework and a transparent diagnostic evidence architecture for future claim-ready evaluation.

## 1. Introduction

Many-to-one DRT systems must coordinate route efficiency, passenger convenience, service reliability, and price while requests arrive sequentially (Cordeau and Laporte, 2007; Ho, Szeto, Kuo, Leung, Petering and Tou, 2018). In a last-mile or feeder setting, a platform can sometimes serve a passenger at home, sometimes ask the passenger to walk to a meeting point, and sometimes decline to serve the request if service is unattractive or infeasible. Meeting points can reduce routing burden and improve consolidation, but they also impose walking and timing burdens on passengers (Stiglic, Agatz, Savelsbergh and Gradisar, 2015; Zheng, Li, Qiu and Wei, 2019; Fielbaum, Bai and Alonso-Mora, 2021; Cortenbach, Gkiotsalitis, van Berkum and Walraven, 2024). These outcomes are shaped not only by routing cost, but also by what the passenger is shown. A displayed alternative is a

service product: it tells the passenger where to be picked up, when pickup is expected, and what price applies.

This paper makes the displayed service menu the central decision object. At request epoch  $i$ , the platform observes the current fleet and route state, generates feasible service bundles, and chooses a limited menu to display. Each displayed bundle has the form  $b = (m, w, p)$ , where  $m$  is the pickup location,  $w$  is the pickup time-window treatment, and  $p$  is the price. The menu also competes with an outside option. If the passenger accepts a home bundle, the request is served as accepted home pickup. If the passenger accepts a meeting-point bundle, the request is served through the selected pickup point. If the outside option is selected, the passenger opts out and the route does not mutate. This separation is essential for transportation operations: accepted service consumes capacity and changes routes, whereas opt-out is lost demand.

 jandxx@buaa.edu.cn (J. Ana)  
ORCID(s):

The paper is written for *Transportation Research Part E* as a service operations and logistics contribution. It connects meeting-point design, pickup-window handling, pricing, passenger choice, route feasibility, and evidence traceability in one dynamic optimization framework. The formulation is not a pricing-only model, a pure route-insertion heuristic, a standalone assortment model, or an attention-model paper. The service menu couples an assortment-style display decision with route-dependent costs and state-dependent feasibility.

The empirical status is deliberately stated at the beginning. The current package is conditional diagnostic, not claim-ready empirical. Phase 4 did not authorize final replay because the pre-replay gate remained blocked by dirty git provenance and missing formal checkpoint evidence. The Phase 10 package reports 74 artifacts, 70 existing artifacts, 4 missing artifacts, and 108 blockers. The strict claim guard allows only provenance/status transparency as a supported claim, while ETA/no-filter material is diagnostic-only. This paper therefore does not convert blocked artifacts into positive results. Instead, it uses the blocker state as a reproducibility boundary: the manuscript can formulate, evaluate diagnostically, audit, and identify conditions for future claim upgrades.

The manuscript makes four claim-safe contributions. First, it formulates a dynamic displayed service-menu problem for many-to-one DRT with bundles  $b = (m, w, p)$ . Second, it provides a self-contained mathematical model covering candidate bundles, displayed menus, MNL response, feasibility constraints, objective components, and separate outcome accounting. Third, it describes an auditable service-menu evaluation pipeline that connects candidate generation, ETA/window handling, pricing, menu selection, replay logging, and claim-gate reporting. Fourth, it documents how

strict claim guards prevent diagnostic, blocked, or scaffold-only material from being used as claim-ready empirical evidence.

## 2. Literature Review

The first stream of related work studies DRT, dial-a-ride, and flexible last-mile service operations. These models usually emphasize routing, capacity, time-window feasibility, fleet repositioning, and service quality (Cordeau and Laporte, 2007; Parragh, Doerner and Hartl, 2008; Ho et al., 2018). Meeting-point DRT extends this setting by allowing passengers to walk to shared pickup points, which can reduce route burden and improve consolidation (Stiglic et al., 2015; Zheng et al., 2019; Fielbaum et al., 2021; Cortenbach et al., 2024). In the present paper, meeting points are not treated as route nodes alone. They are components of displayed service bundles, and their operational value depends on time-window treatment, price, and passenger response.

The second stream concerns pickup-window design and service reliability. Time-window promises affect both passenger acceptance and operational feasibility. A narrow pickup window may make the service more attractive but harder to route; a flexible window may reduce route pressure but lower perceived convenience. Robust ETA filtering and adaptive-window logic are therefore natural service-menu ingredients. Under the current evidence state, however, these ingredients are model components and diagnostic comparison slots rather than authorized effect claims.

The third stream is assortment and menu optimization under discrete choice. Retail assortment models choose which products to offer when customers may substitute or choose an outside option (Talluri and van Ryzin, 2004; Rusmevichientong, Shen and Shmoys, 2010; Desir, Goyal and Zhang, 2022). A DRT service menu has a related

choice structure, but each product is state-dependent and operationally endogenous. Passenger response is represented with a multinomial-logit structure grounded in the discrete-choice literature (McFadden, 1974; Ben-Akiva and Lerman, 1985). The cost of a bundle depends on the current route, capacity, pickup location, time window, and feasibility checks. The menu decision is therefore both a display decision and a transportation operations decision.

The fourth stream integrates pricing with operational decision-making. Pricing can shape willingness to accept meeting-point service or time-window trade-offs. In this paper, Lambert-W pricing is one price-generation component inside a wider service-menu pipeline. It does not define the full contribution by itself. The paper also separates price generation from menu feasibility, choice response, route mutation, and artifact-gated evidence interpretation.

Finally, computational transportation studies increasingly need transparent provenance and reproducibility controls. Decision-focused and predict-then-optimize pipelines underline why predictive and optimization components should be evaluated through the downstream decision they support (Elmachtoub and Grigas, 2022). The current manuscript treats paired replay, checkpoint load status, normalized rows, artifact status, package status, and strict claim guards as first-class evidence controls. This framing does not replace empirical evidence. It prevents a manuscript from overstating what the generated evidence can support and clarifies what future work must regenerate before stronger claims are available.

### 3. Problem Description

Requests arrive sequentially. At epoch  $i$ , a passenger request is observed together with the current system state  $S_i$ .

The state includes the active fleet and route plan, vehicle capacity, time information, the passenger's home location, candidate meeting points, routing and insertion costs, pickup-window metadata, price settings, and service guardrails. The platform must decide which service alternatives to display before the passenger response is known.

A service alternative is a bundle  $b = (m, w, p)$ . The pickup component  $m$  may be the passenger's home or a feasible meeting point. The time-window component  $w$  records how pickup timing is offered, such as fixed-window or adaptive-window handling. The price component  $p$  is the displayed fare or discounted fare associated with the bundle. A displayed menu  $M_i$  is a subset of feasible bundles and is constrained by a menu-size limit.

The passenger may choose one displayed bundle or choose the outside option. The outside option is refusal or lost demand. It is not home service and should not change the route. This distinction yields three accounting outcomes. Accepted home pickup means the passenger chooses a home-service bundle and the vehicle route is updated for home pickup. Accepted meeting-point pickup means the passenger chooses a meeting-point bundle and the route is updated for that location. Opt-out means the passenger chooses the outside option and no pickup route is inserted.

The operator's decision is difficult because passenger attractiveness and operational cost move together. A far meeting point may reduce vehicle cost but reduce passenger utility. A home pickup may be attractive but costly. A price can compensate or discourage acceptance. A pickup window can increase reliability for one request but reduce downstream flexibility. Dynamic service-menu optimization evaluates these trade-offs under current route state, feasible products, and evidence gates that determine which results may be interpreted as claim-bearing.

#### 4. Mathematical Model

Let  $\mathcal{I}$  be the sequence of requests. For request  $i \in \mathcal{I}$ , let  $S_i$  denote the system state before the menu is displayed. Let  $L_i$  be the feasible pickup-location set, including home service and any feasible meeting-point alternatives. For each  $m \in L_i$ , let  $W_i(m)$  be the feasible set of pickup-window treatments, and let  $P_i(m, w)$  be the feasible set of prices for location-window pair  $(m, w)$ .

The candidate bundle set is

$$B_i = \{b = (m, w, p) : m \in L_i, w \in W_i(m), p \in P_i(m, w), b \in \mathcal{F}_i(S_i)\}, \quad (1)$$

where  $\mathcal{F}_i(S_i)$  is the set of route-, capacity-, ETA/window-, and guardrail-feasible bundles under state  $S_i$ . The platform chooses binary display variables  $x_{ib}$ , where  $x_{ib} = 1$  if bundle  $b$  is displayed. The menu is

$$M_i = \{b \in B_i : x_{ib} = 1\}, \quad (2)$$

with menu-size constraint

$$\sum_{b \in B_i} x_{ib} \leq K. \quad (3)$$

The feasibility constraints include the following conditions. Pickup-location feasibility requires the location to be home service or an available meeting point for request  $i$ . ETA/window feasibility requires the predicted pickup time to be compatible with the bundle's time-window treatment. Capacity and route feasibility require the insertion to respect vehicle capacity and route state. Service guardrails exclude alternatives that violate the service contract. Policy-comparison comparability requires paired replay settings, seeds, demand splits, checkpoint requirements, and policy-only overrides to remain controlled across policies.

Passenger response is represented by an MNL model over displayed bundles and the outside option. Let  $U_{ib}$  be the systematic utility of displayed bundle  $b$ :

$$U_{ib} = \alpha_0 + \alpha_p p_{ib} + \alpha_d d_{ib} + \alpha_t t_{ib} + \alpha_w q_{ib} + \alpha_h h_{ib}, \quad (4)$$

where  $d_{ib}$  is walking distance,  $t_{ib}$  is pickup-time or in-vehicle-time burden,  $q_{ib}$  summarizes time-window compatibility, and  $h_{ib}$  indicates home service. Let  $U_{i0}$  denote outside-option utility. The probability of choosing displayed bundle  $b$  is

$$\Pr_i(b \mid M_i) = \frac{\exp(U_{ib})}{\exp(U_{i0}) + \sum_{j \in M_i} \exp(U_{ij})}. \quad (5)$$

The outside-option probability is

$$\Pr_i(0 \mid M_i) = \frac{\exp(U_{i0})}{\exp(U_{i0}) + \sum_{j \in M_i} \exp(U_{ij})}. \quad (6)$$

The expected one-request menu objective can be written

as

$$\begin{aligned} \max_x \quad & \sum_{b \in B_i} x_{ib} \Pr_i(b \mid M_i) [p_{ib} - c_{ib}(S_i)] \\ & - \lambda_{\text{out}} \Pr_i(0 \mid M_i) - \Phi(M_i, S_i), \end{aligned} \quad (7)$$

where  $c_{ib}(S_i)$  is predicted operational cost,  $\lambda_{\text{out}}$  is an opt-out/lost-demand penalty, and  $\Phi(M_i, S_i)$  collects service-quality, ETA/window, and guardrail penalties. This representation is intentionally general: concrete runtime policies can use different menu-objective modes, pricing modes, and solver contracts while preserving the same evidence accounting.

The realized outcome after choice is recorded separately. If the chosen bundle is home service, the row increments accepted home pickup. If the chosen bundle is a meeting-point service, the row increments accepted meeting-point

**Table 1**

Diagnostic service-menu evaluation pipeline.

| Step | Operation                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Read current route, capacity, request, and replay state.                                                                                                                                                                                                                                                    |
| 2    | Generate candidate pickup locations: home plus feasible meeting points.                                                                                                                                                                                                                                     |
| 3    | For each location, generate candidate pickup-window treatments.                                                                                                                                                                                                                                             |
| 4    | For each location-window pair, generate or assign prices.                                                                                                                                                                                                                                                   |
| 5    | Form candidate bundles $b = (m, w, p)$ .                                                                                                                                                                                                                                                                    |
| 6    | Apply route, capacity, ETA/window, and service-guardrail filters.                                                                                                                                                                                                                                           |
| 7    | Select a displayed menu subject to menu-size and policy contracts. Use exact enumeration when candidate count is inside the exact limit; use greedy fallback when candidate count exceeds that limit; record solver diagnostics, candidate counts, fallback reason, build time, and selected menu metadata. |
| 8    | Evaluate the MNL response model over displayed bundles plus outside option.                                                                                                                                                                                                                                 |
| 9    | Apply the selected outcome: accepted home pickup mutates the route through home service; accepted meeting-point pickup mutates the route through the chosen meeting point; opt-out records lost demand and does not mutate the route.                                                                       |
| 10   | Emit normalized row fields, checkpoint load status, policy tag, method metadata, manifest/settings hashes, and outcome counts.                                                                                                                                                                              |
| 11   | Classify generated artifacts through readiness, row, artifact, package, and strict claim-guard gates.                                                                                                                                                                                                       |

pickup. If the outside option is chosen, the row increments opt-out and the route is not mutated. Generated rows also record checkpoint status, policy tag, settings hash, manifest hash, outside-option utility, and method metadata. These fields are required because formal interpretation depends on provenance and artifact gates, not only on numerical objective values.

## 5. Solution Method

The method is an online diagnostic service-menu pipeline. It uses the current request and route state to generate feasible products, score menus, record choice and outcome metadata, and route the resulting evidence through claim gates. The pipeline is written as pseudocode because the manuscript's purpose is to describe an auditable evaluation contract, not to assert claim-ready algorithmic performance.

Candidate bundle generation starts from feasible service products. Home service may be retained as a displayed option or may be part of the candidate pool, depending on the policy contract. Meeting-point candidates carry walking distance, predicted route cost, ETA, time-window metadata,

and service-quality features. The service guardrails prevent infeasible or contract-violating alternatives from being displayed.

ETA and pickup-window handling are modeled as feasibility and scoring components. Robust ETA filters can exclude or penalize bundles whose predicted pickup timing is outside the relevant service window. A no-filter diagnostic path can expose how the ETA filter changes the candidate set, but it is not an operating recommendation in this manuscript. Fixed-window and adaptive-window settings are comparison slots governed by the strict claim guard.

Pricing is applied after candidate construction or during menu evaluation. The Lambert-W pricing component maps utility and cost terms into candidate prices when that pricing mode is active. The paper treats this component as part of a larger service-menu method, because the displayed decision also depends on route feasibility, menu size, choice response, outside-option treatment, and claim-gated evidence status.

Menu selection may use exact enumeration for small candidate sets and greedy fallback for larger sets. Current computational material is diagnostic and C6 is blocked, so the method section does not assert computational credibility.

**Table 2**

Primary policy family and paired-comparison roles.

| Policy tag                      | Role in paired comparison                                 |
|---------------------------------|-----------------------------------------------------------|
| mainline_no_menu                | Baseline slot without displayed service-menu construction |
| mainline_fixed_menu             | Fixed service-menu slot                                   |
| mainline_random_menu            | Random displayed-menu diagnostic slot                     |
| mainline_optimized_m            | Optimized meeting-point-only product slot                 |
| mainline_optimized_mw           | Optimized meeting-point and window product slot           |
| mainline_optimized_fixed_window | Optimized menu with fixed-window handling                 |
| mainline_optimized_adaptive     | Optimized menu with adaptive-window handling              |

It reports the contract: the runtime records the effective solver, threshold, candidate count, fallback flag, fallback reason, and related diagnostics so future evidence can be audited.

## 6. Experimental Design

The evaluation design is paired replay. Each policy comparison must preserve the same demand splits, seeds, replay settings, checkpoint requirements, manifest structure, and non-policy runtime settings. Only declared policy overrides may differ. This design prevents a comparison from mixing policy effects with changes in scenario, checkpoint provenance, or replay conditions.

These tags define the comparison surface. Under the current strict claim guard, they do not authorize directional effect language. They are used to explain how the service-menu family is organized and what future claim-ready evidence would need to compare.

The normalized rows record profit and service metrics, including `net_profit`, `acceptance_rate`, `optout_rate`, `accepted_home_count`, `accepted_meeting-point_count`, `home_share`, `meeting-point_uptake`, `served_rate`, and `status` fields. Provenance fields include `policy_tag`, `manifest_hash`, `settings_hash`, `checkpoint_path`, `checkpoint_hash`, `checkpoint_requirement`, and `checkpoint_load_status`. The checkpoint fields are not administrative details: formal and pilot evidence cannot

support manuscript claims without explicit load status and matching metadata.

Evidence is classified into four tiers. Formal evidence can support claim-bearing language only when readiness, row validity, artifact status, package status, and strict claim guard all pass. Diagnostic evidence can support mechanism and boundary discussion. Blocked evidence can support status/provenance transparency but not positive empirical conclusions. Scaffold-only material can define future study protocols but cannot establish executed case evidence. The claim gates are therefore part of the experimental design.

## 7. Results

The Results section begins with the strict claim gate. In the current package, the claim-ready flag, strict claim-guard flag, and positive-claims permission flag are all false. It contains 74 package artifacts, 70 existing artifacts, 4 missing artifacts, and 108 blockers. The package is usable for diagnostic interpretation and provenance transparency, but it is not claim-ready empirical evidence.

This claim table shapes the diagnostic findings. First, the service-menu framework can be described as implemented and auditable, but central adaptive menu performance claims remain blocked. Second, product and time-window ablations can be described as diagnostic comparison structure, not as claim-ready evidence of value. Third, ETA/no-filter material

**Table 3**

Claim status and allowed manuscript use under the current strict guard.

| Claim                | Current status      | Allowed manuscript use                                    |
|----------------------|---------------------|-----------------------------------------------------------|
| C1 adaptive menu     | unsupported blocked | Not allowed as a positive claim; status and blockers only |
| C2 product ablation  | diagnostic blocked  | Diagnostic structure and future upgrade conditions only   |
| C3 adaptive window   | unsupported         | Comparison slot only; no directional effect language      |
| C4 menu construction | diagnostic blocked  | Auditable mechanism discussion only                       |
| C5 ETA boundary      | diagnostic only     | ETA/no-filter boundary discussion only                    |
| C6 exact/greedy      | blocked diagnostic  | Computational diagnostic appendix only                    |
| C7 provenance        | status supported    | Provenance/status transparency only                       |
| C8 semi-real case    | scaffold blocked    | Future-study scaffold only                                |

can identify a robustness boundary, but no-filter is not presented as a deployment rule. Fourth, exact/greedy output can document computational status and fallback metadata, while computational credibility remains blocked. Fifth, case-study material remains scaffold-only and future-study oriented.

The source-family status reinforces this interpretation. Main RC artifacts are blocked because formal readiness and checkpoint evidence are unavailable. Phase 8 sensitivity artifacts are diagnostic and provisional. Phase 9 tractability artifacts are diagnostic and provisional. Case material is scaffold-only. Blocker-status documents are provenance/status material. The result is a transparent diagnostic package rather than a positive empirical result package.

This interpretation is not a paper weakness to hide; it is the paper's evidence discipline. A reader can see which model components exist, which comparison slots are planned, which generated artifacts are blocked, which diagnostic boundaries are visible, and what future regenerated evidence would need to change before any claim ID is upgraded.

## 8. Discussion

The primary contribution is the formulation of a displayed service-menu decision for many-to-one DRT. The model ties meeting points, pickup-window handling,

prices, MNL response, route feasibility, and outside-option accounting into one sequential decision. That contribution is useful even under a diagnostic evidence ceiling because it clarifies what the platform actually controls and what generated evidence must preserve.

The second contribution is the evidence architecture. Paired replay controls scenario comparability. Checkpoint metadata records whether required prediction models were available and loaded. Normalized rows separate accepted home pickup, accepted meeting-point pickup, and opt-out. Artifact gates classify evidence into formal, diagnostic, blocked, and scaffold-only tiers. Strict claim guards then translate those tiers into allowed manuscript language. This makes the manuscript harder to overstate and easier to audit.

The limitation is equally explicit. Final replay was not run in this milestone because the pre-replay gate remained blocked. The current package has `claim_ready=false`, and the guard blocks positive claims for C1, C2, C3, C4, C6, and C8. The manuscript therefore cannot conclude that one menu family has claim-ready performance effects over another. It can only state the formulation, diagnostic boundaries, blocker status, and future evidence path.

The reviewer implications are direct. If a reviewer asks where the adaptive menu performance claim is, the answer is that C1 is blocked and not asserted. If a reviewer asks whether no-filter should be used operationally, the answer

is that C5 is diagnostic boundary material only. If a reviewer asks whether case material reflects executed passenger evidence, the manuscript explicitly does not claim such behavior because C8 remains scaffold-only. If a reviewer asks whether exact/greedy material establishes solver quality, the answer is that C6 is blocked diagnostic and future evidence is required.

Future claim upgrades require new evidence, not editorial substitution. A future milestone would need clean provenance, loaded checkpoint status and sidecar metadata, complete formal rows, dependency snapshots, artifact-status passes, package regeneration, and a strict claim guard that authorizes the exact claim ID. Only then should the manuscript wording change from conditional diagnostic to claim-ready empirical.

## 9. Conclusion

This paper formulates dynamic displayed service-menu optimization for many-to-one DRT. The central decision is the displayed bundle  $b = (m, w, p)$ , combining pickup location, pickup-window handling, and price. The model defines candidate bundles, menus, MNL response, objective components, feasibility constraints, outside option, accepted home pickup, accepted meeting-point pickup, and opt-out accounting. The method describes an auditable diagnostic pipeline from candidate generation through claim-gate reporting.

The current manuscript remains conditional diagnostic with `claim_ready=false`. It provides a service-menu formulation, paired-replay evaluation architecture, and transparent claim-boundary audit. Positive empirical claims remain blocked by readiness and strict claim-guard status, and final replay was not authorized in the current milestone. The

paper is therefore a claim-gated diagnostic manuscript and a foundation for future clean evidence regeneration.

## A. Source Map And Claim Audit

The manuscript uses `manuscript/TR_E_WORK2_TABLE_FIGURE_SOURCE_MAP.md` as the source map for planned manuscript objects. Each object is required to identify source artifact path, claim ID, claim status, allowed manuscript use, and evidence class. The strict claim boundary is documented in `manuscript/TR_E_WORK2_CLAIM_AUDIT.md` and the generated `work2_coding/artifacts/work2_robust_menu/phase10_paper_artifacts/CLAIM_GUARD.json`.

## B. ETA And No-Filter Diagnostic Boundary

ETA/no-filter material belongs to `C5_eta_robustness_boundary` and is diagnostic-only under the current guard. It may be used to identify boundary conditions created by ETA filtering and pickup-window handling. It does not authorize a no-filter operating recommendation or any deployment language.

## C. Exact/Greedy Computational Diagnostics

Exact enumeration and greedy fallback are runtime contracts whose diagnostics include candidate count, effective solver, threshold, fallback flag, fallback reason, and build time. Current Phase 9 material is diagnostic and provisional, and `C6_exact_greedy_computational_credibility` remains blocked. The appendix may report computational status only within that boundary.

## D. Case-Scaffold Boundary

Case-study material under `.planning/data/case_studies/` is scaffold-only. It may define future source contracts, route-selection ideas, and demand protocols. It cannot be presented



as executed case evidence, real passenger evidence, or claim-ready validation in the current manuscript.

## E. Prohibited-Language Check

The revised prohibited-language check is maintained in manuscript/TR\_E\_WORK2\_REVISIED\_PROHIBITED\_LANGUAGE\_CHECK.md. Hits from the scan must be classified as safe blocked-claim/status discussion, safe explicit denial, or unsafe language requiring manuscript revision. The final status is pass only when no unqualified positive claim remains.

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